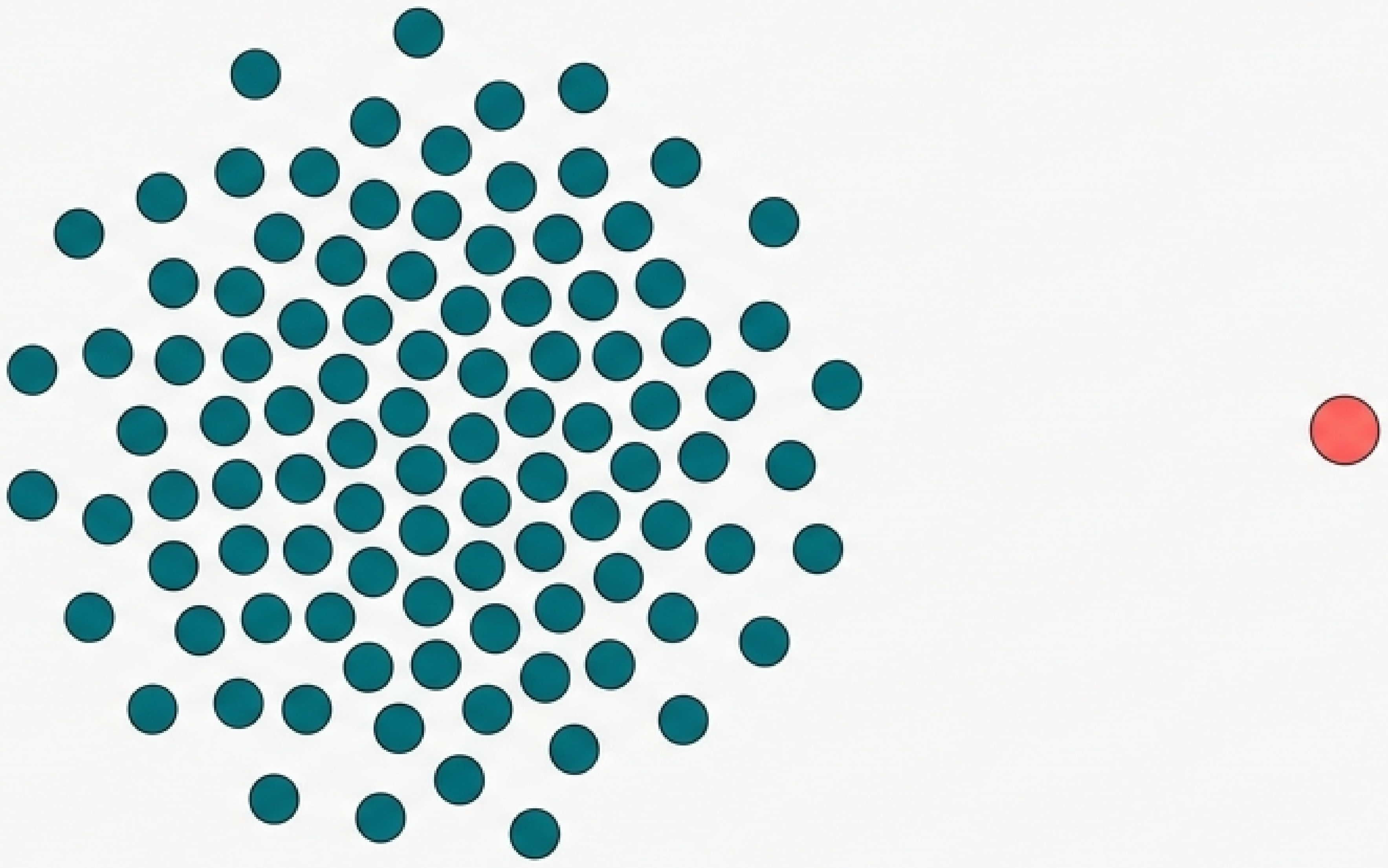
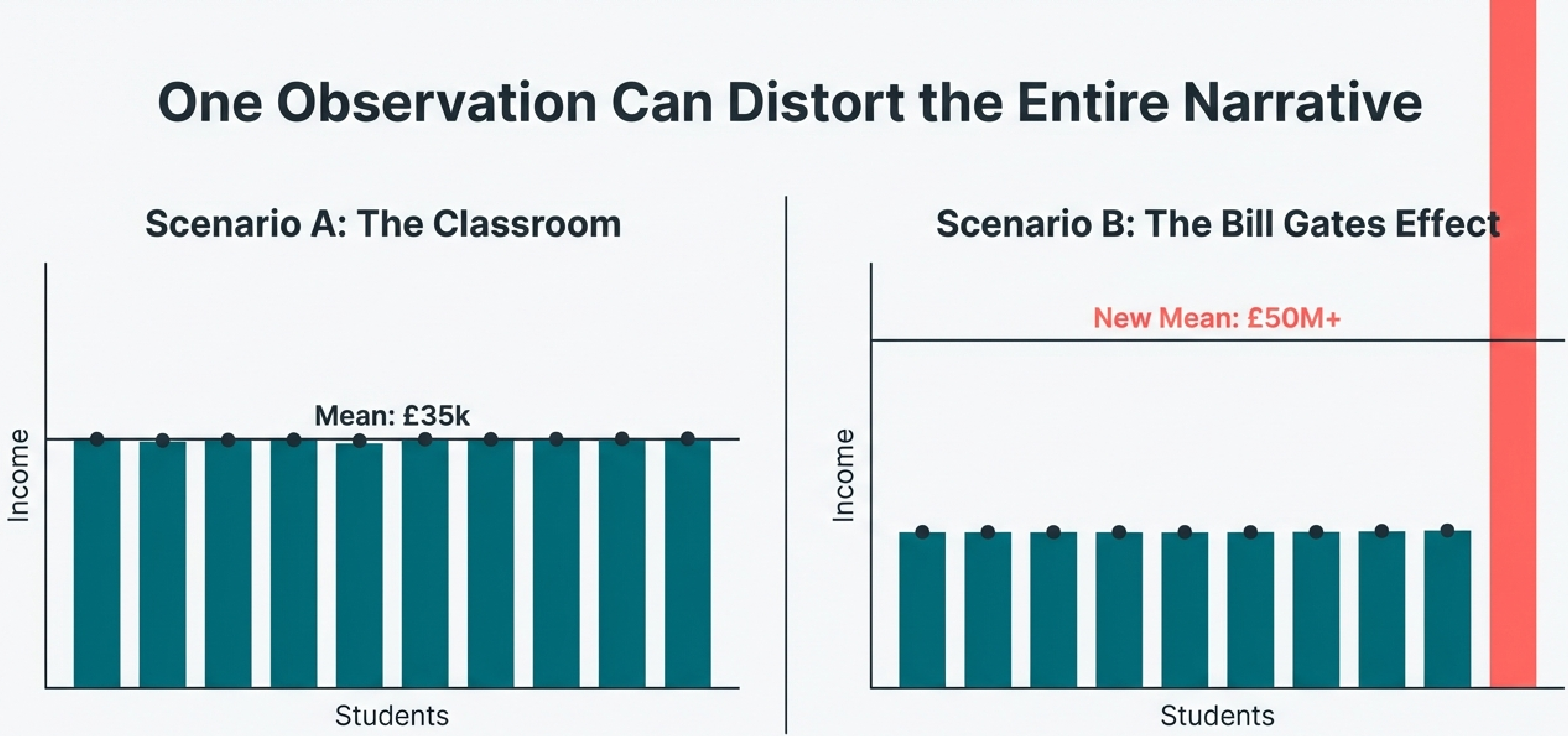
Mastering Outliers in Feature Engineering

A diagnostic framework for detection, impact assessment, and treatment strategies in machine learning.



Based on the lecture series by CampusX

One Observation Can Distort the Entire Narrative

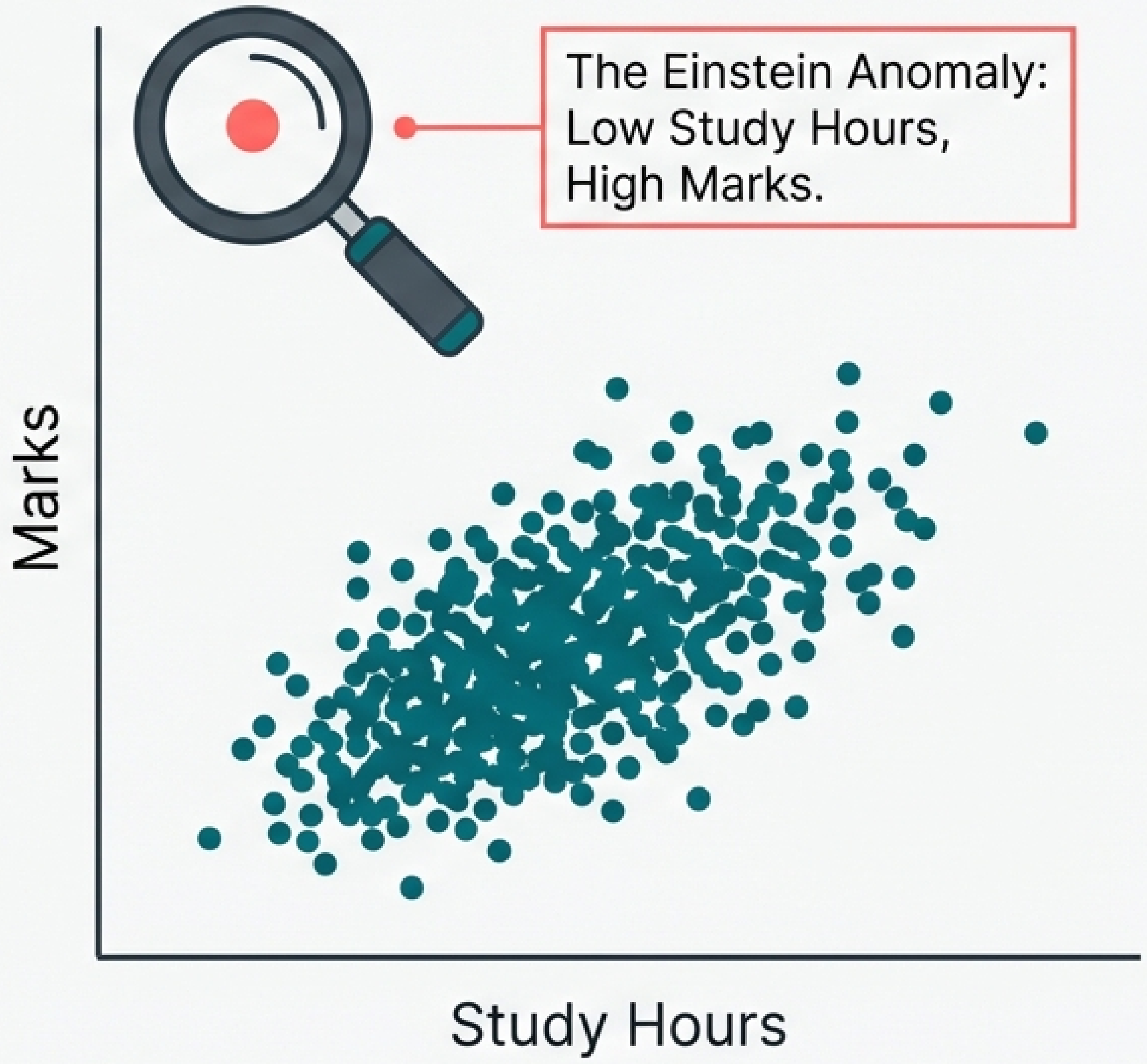


Intuition: When Bill Gates enters a room of average earners, the mean income suggests everyone is a millionaire. The outlier shifts the central tendency, misleading the model about the ‘typical’ behaviour of the group.

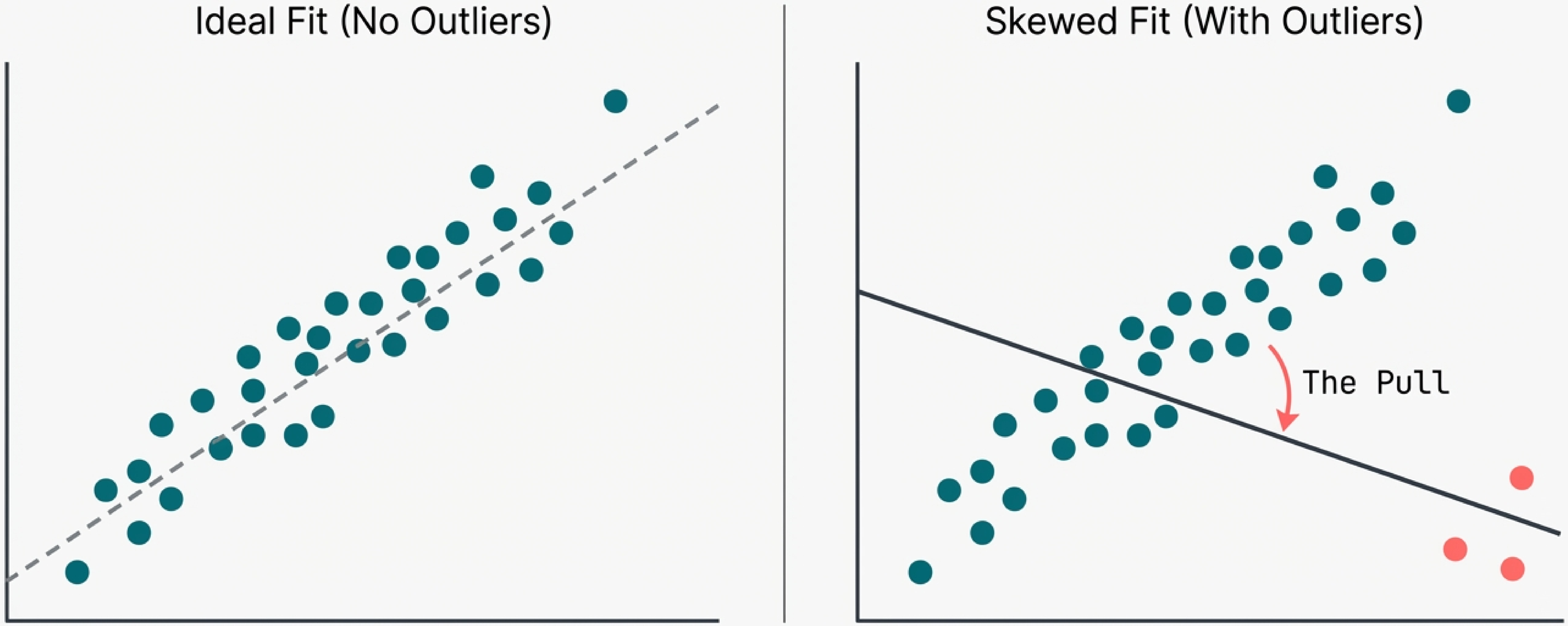
Defining the Outlier

Definition: An observation that deviates so significantly from other observations as to arouse suspicions that it was generated by a different mechanism.

The Distraction Factor: Machine Learning models aim to find generalisable patterns. Outliers force the model to over-accommodate exceptions, shifting focus away from the majority trend.



The Tug of War: How Outliers Skew Relationships

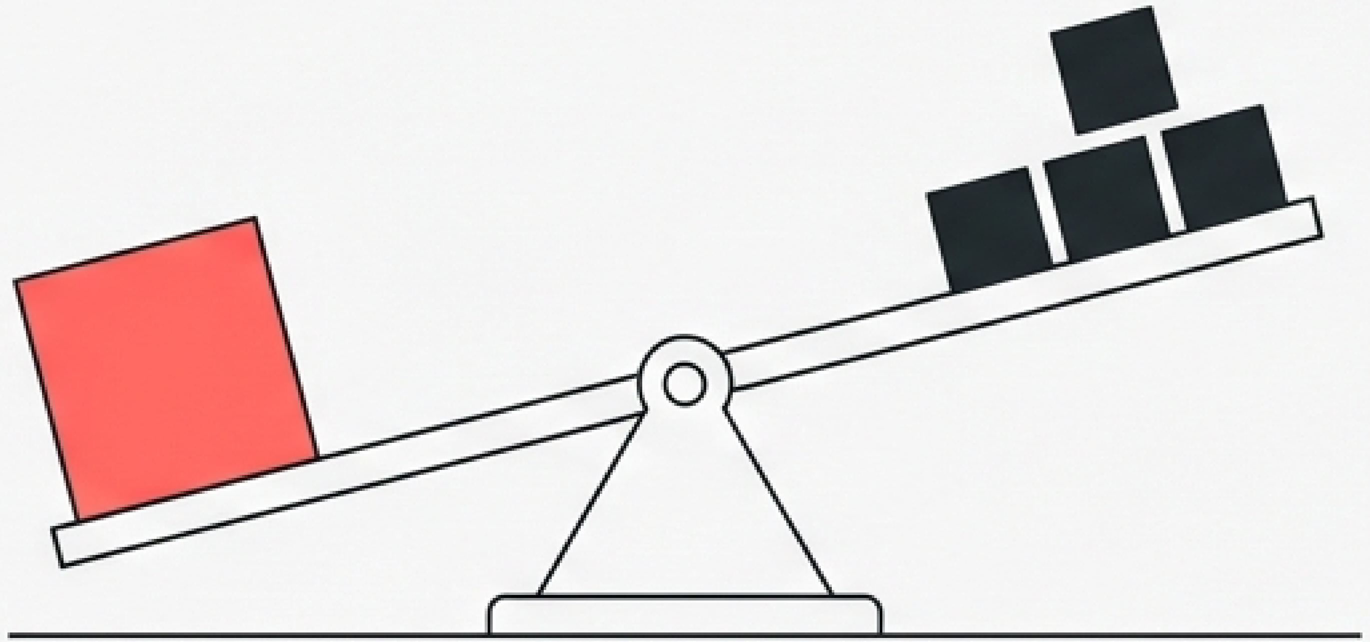


Impact: To minimise error distance (MSE) for the outlier, the algorithm sacrifices accuracy for the majority. The model 'fails' the normal students to accommodate the anomaly.

Algorithm Sensitivity Protocol

High Sensitivity (Must Treat)

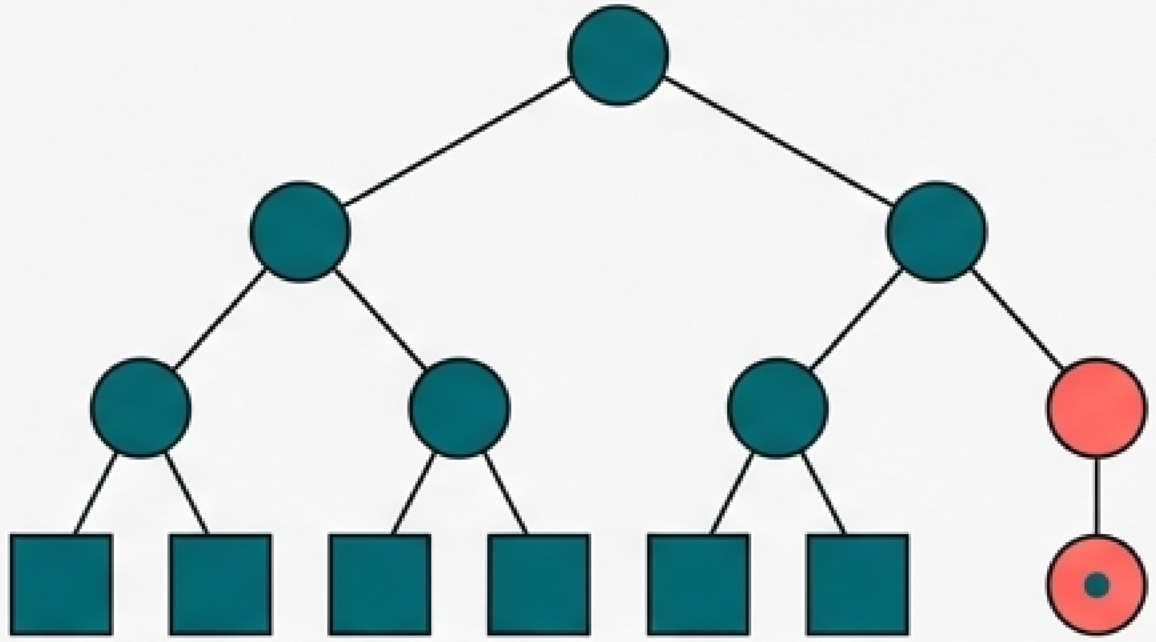
- Linear & Logistic Regression
- Deep Learning (Neural Networks)
- Adaboost



Weight-Based: These models calculate coefficients. Large errors cause drastic weight updates.

Low Sensitivity (Robust)

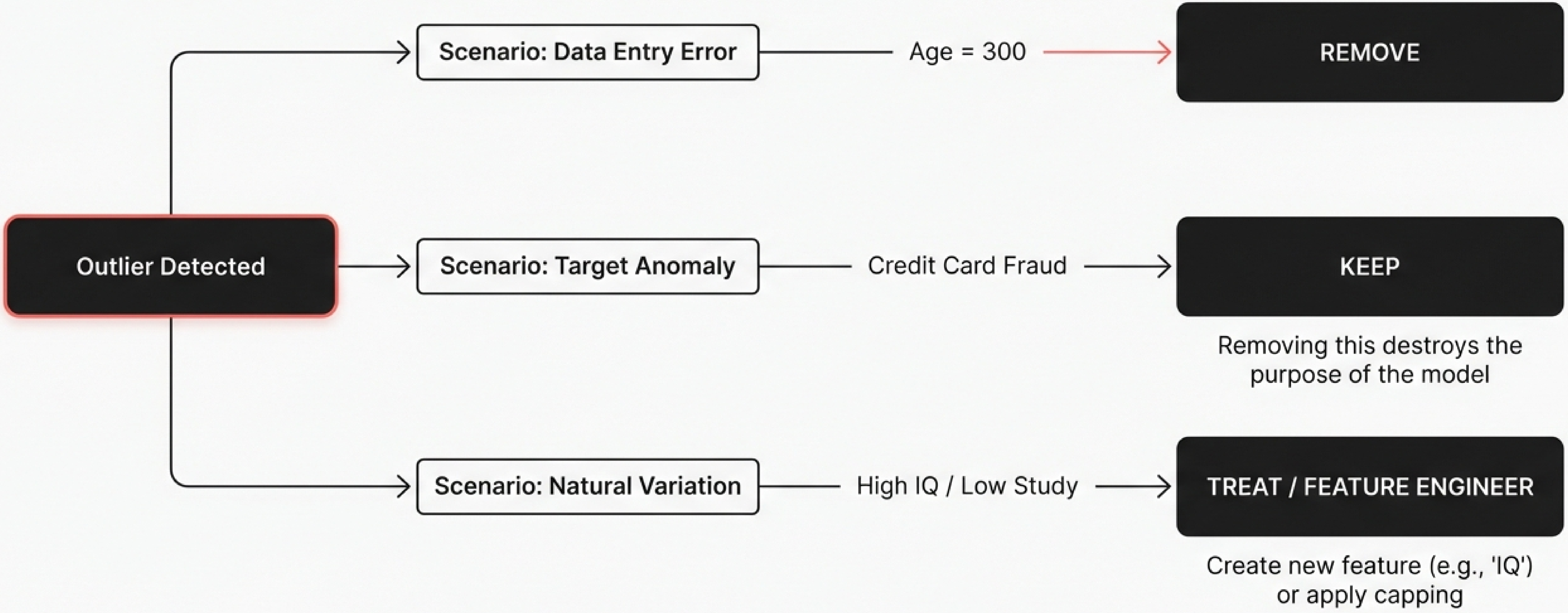
- Random Forest
- XGBoost
- Decision Trees



Space-Partitioning: These models “cut” the data. An outlier is simply isolated in its own leaf node without shifting boundaries for others.

Clinical Editorial Blueprint

The Decision Gate: Error, Anomaly, or Feature?



NotebookLM

Detection Protocols

The method of detection depends on the shape of the data distribution.

Z-Score
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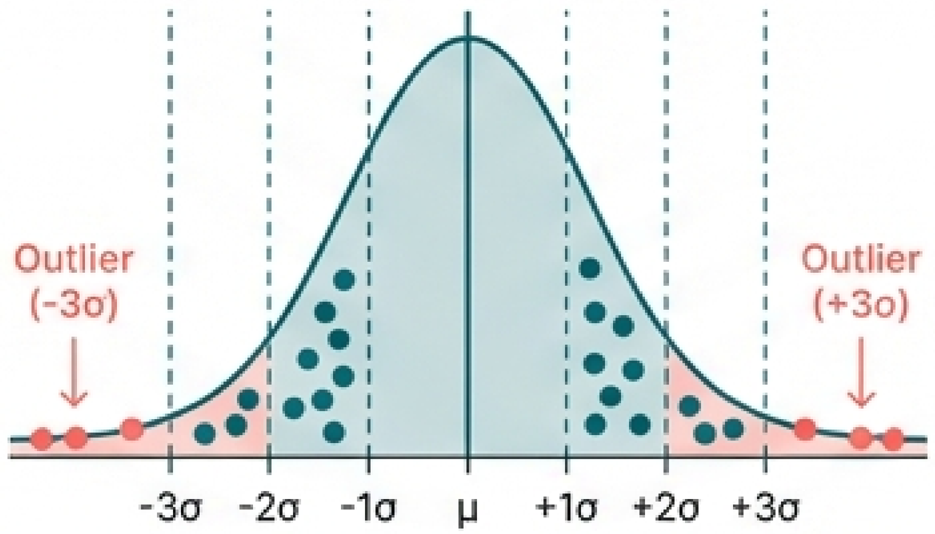

For Normal Distributions
Inter Regular

IQR Method
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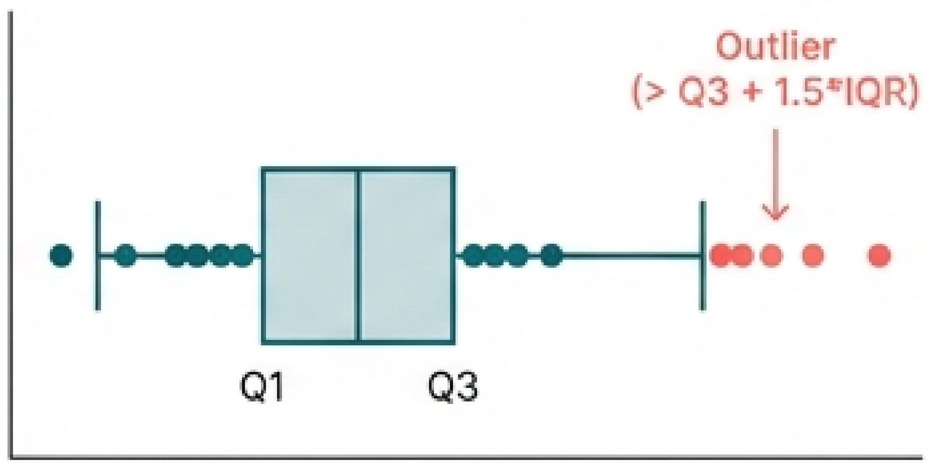

For Skewed Distributions
Inter Regular

Percentile
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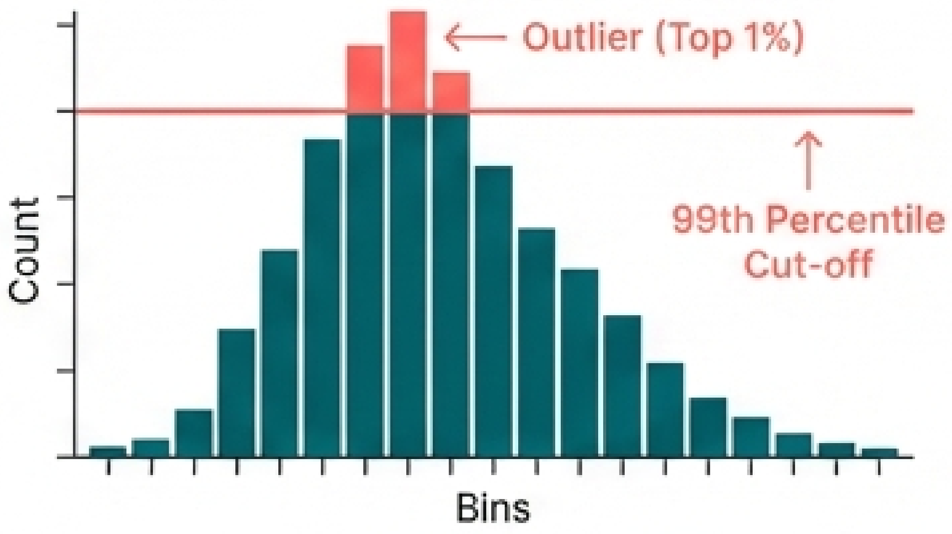
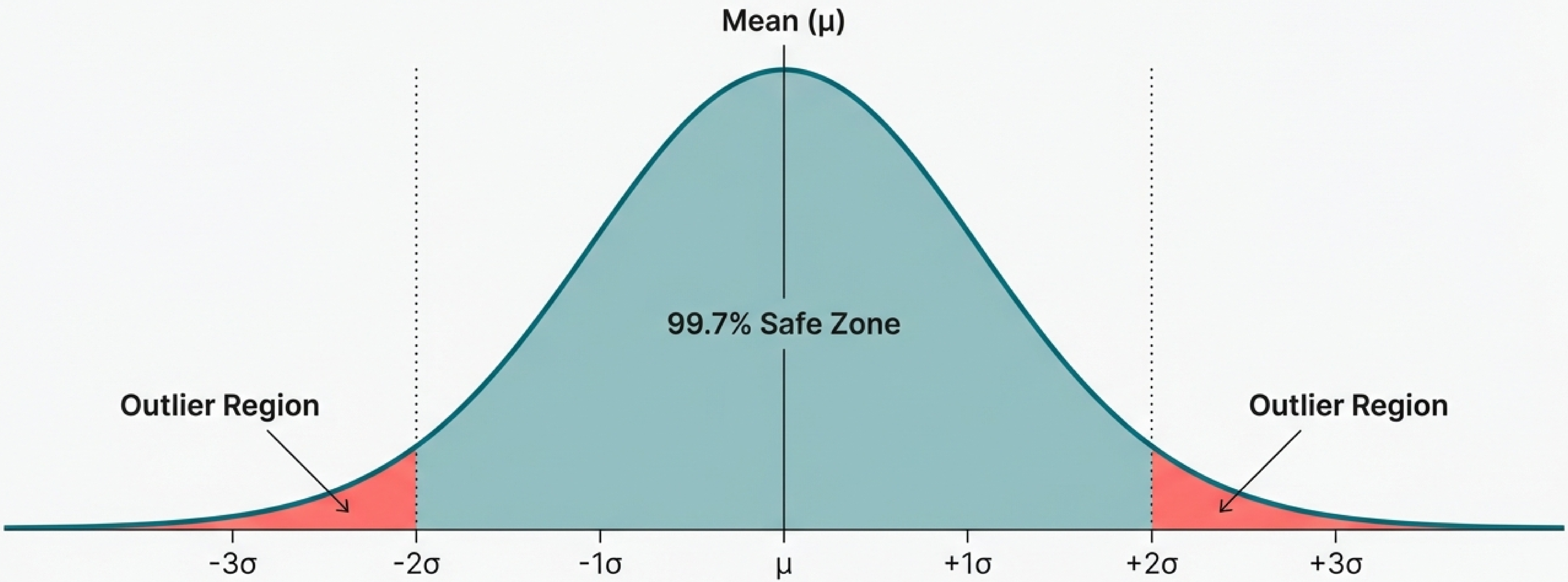


For Unknown / Arbitrary Thresholds
Inter Regular

Protocol 1: The Z-Score

Standard Deviation approach for Normal Distributions.



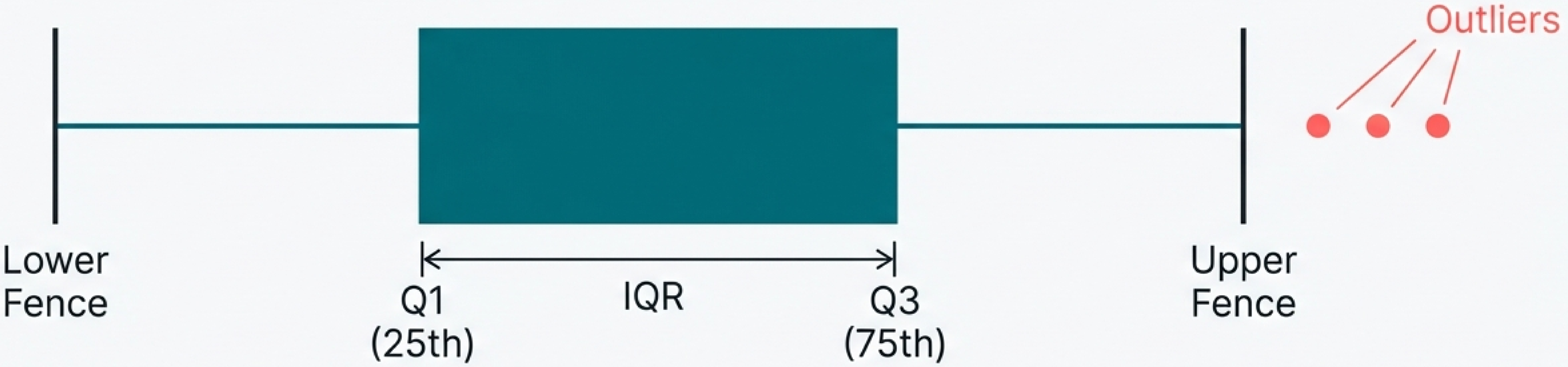
The Logic: In a Gaussian distribution, 99.7% of data lies within 3 standard deviations of the mean.

```
Upper Limit = Mean + 3(SD)
Lower Limit = Mean - 3(SD)
```

Clinical Editorial Blueprint

Protocol 2: IQR Proximity Rule

Box Plot approach for Skewed Data.




```
IQR = Q3 - Q1
Lower Fence = Q1 - (1.5 * IQR)
Upper Fence = Q3 + (1.5 * IQR)
Note: The 1.5 multiplier determines the sensitivity of the fence.
```

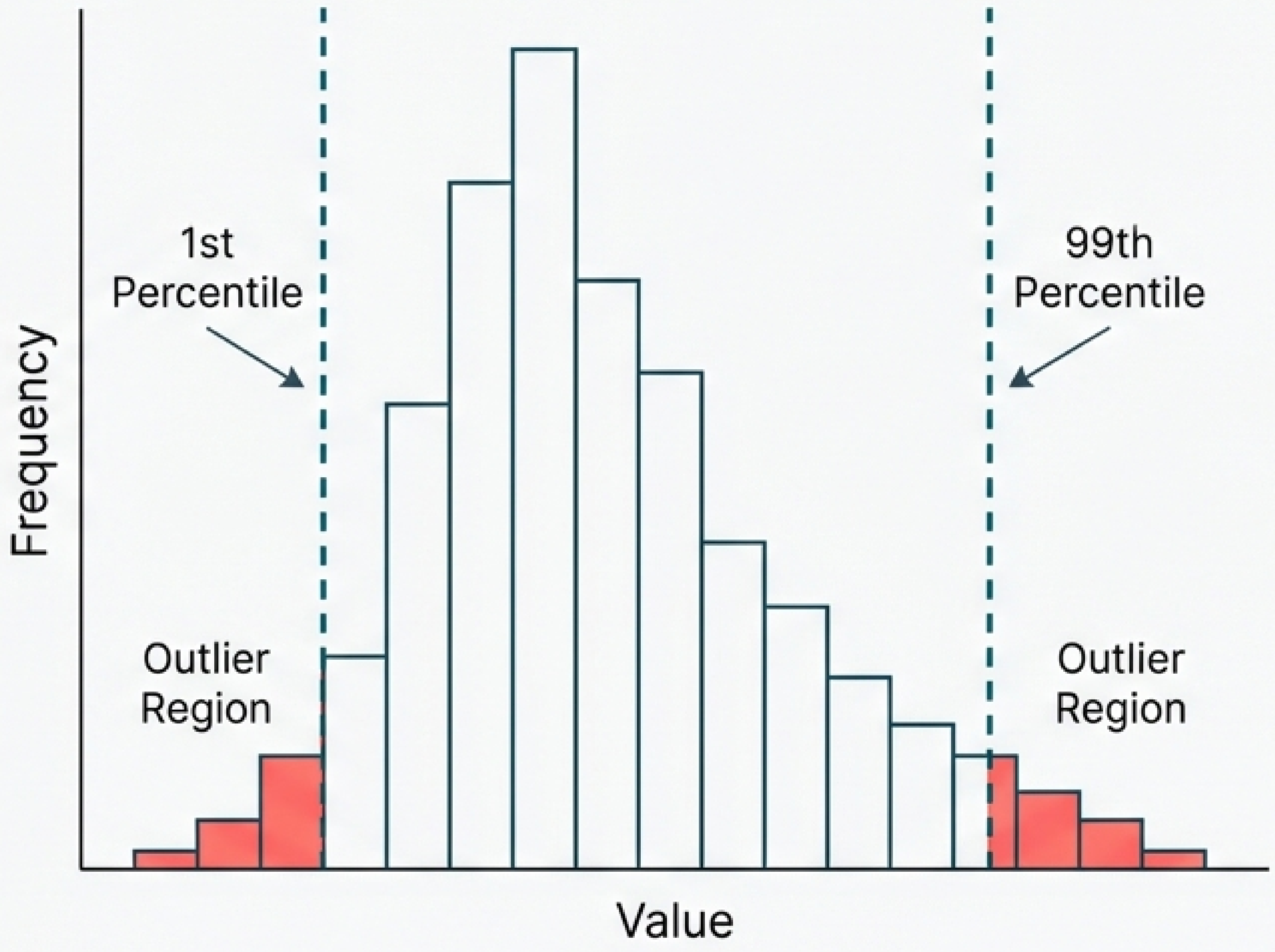
Protocol 3: Percentile Capping

When distribution properties are unknown, apply hard thresholds based on volume.

Logic:
When distribution properties are unknown, apply hard thresholds based on volume.

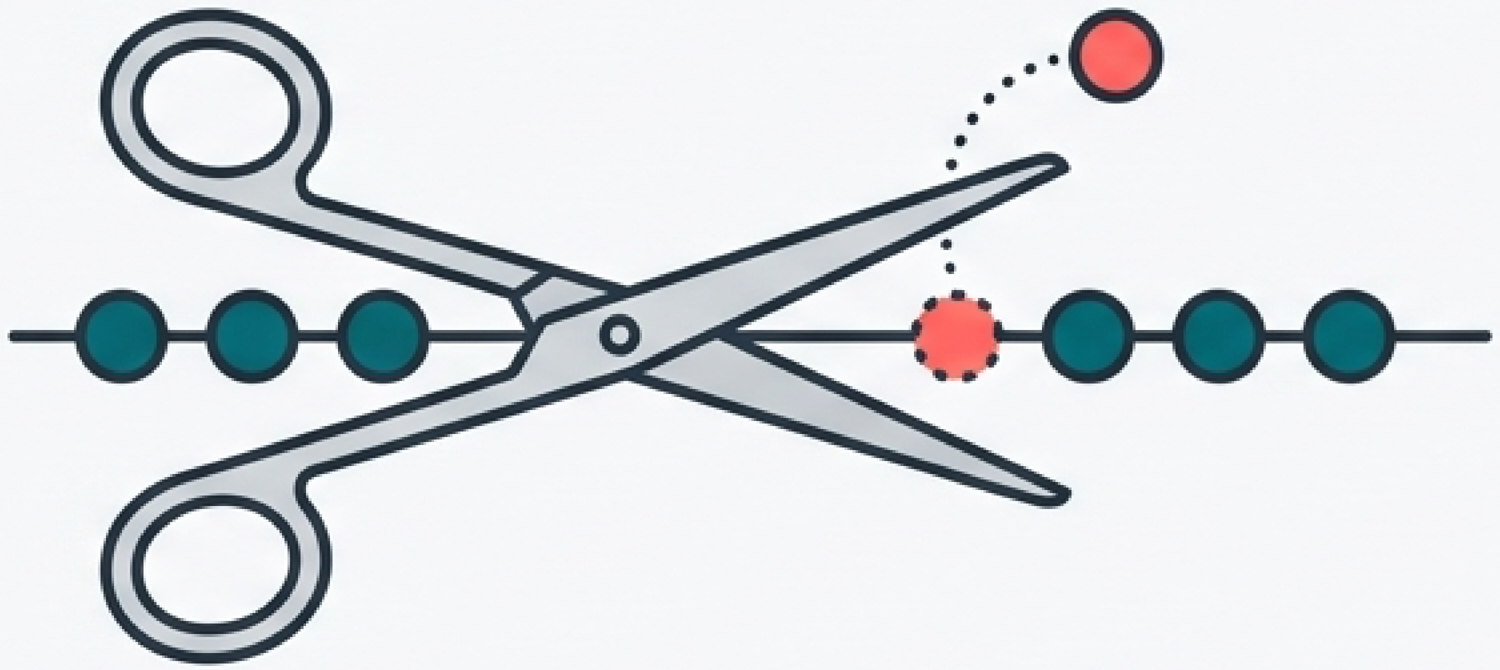
Method:
Flag any values exceeding the 99th percentile or falling below the 1st percentile.

Hyperparameter:
The threshold (e.g., `99` vs `97.5`) is chosen based on domain knowledge.



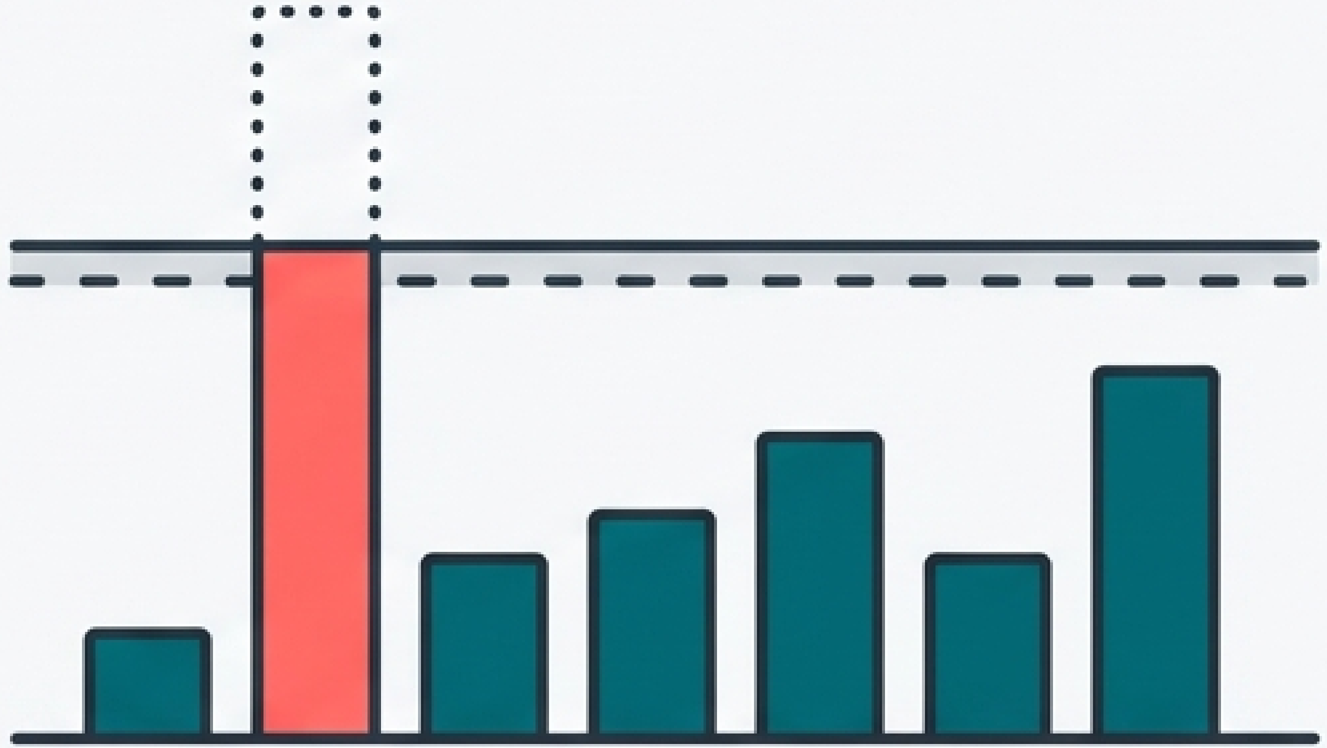
Treatment Strategies: Cure or Control?

Strategy A: Trimming



The Surgery. Complete removal of the outlier observations from the dataset.

Strategy B: Capping (Winsorization)



The Limit. Modifying extreme values to align with the nearest safe threshold.

Alternative (Minor): Treat as Missing Values (Discretisation).

Technique A: Trimming

Mechanism: Exclude outlier rows entirely.

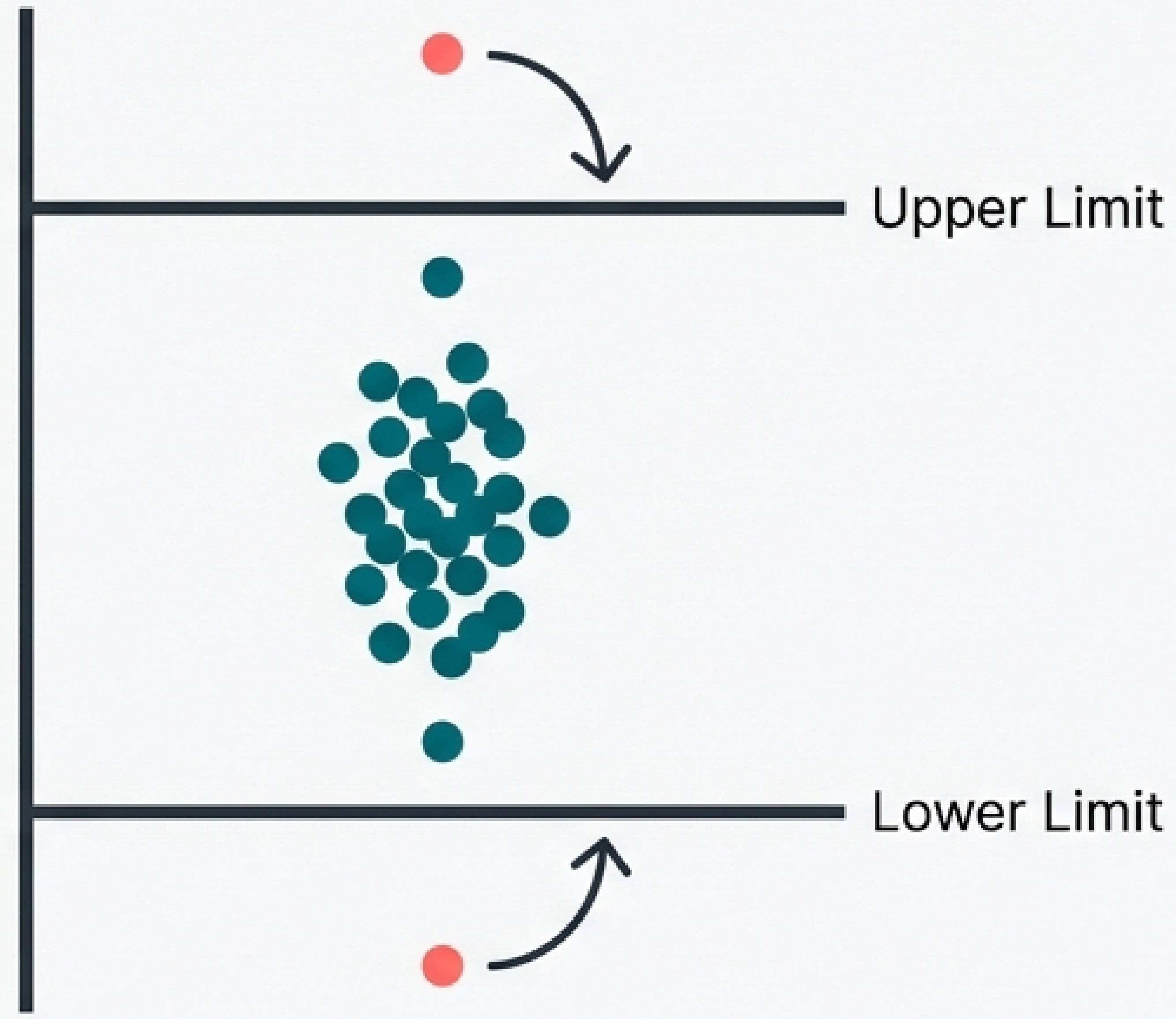
Best For: Large datasets with few, extreme errors (e.g., 'Age 300').

Pros: Fast, conceptually simple. Removes noise completely.

Cons: Reduces data size. Risk of information loss in small datasets.

Technique B: Capping (Winsorization)

- **Mechanism:** Replace extreme values with the calculated threshold limits.
- **Logic:** Outliers exist at the extremes. By capping, we preserve the record but neutralise the skew.
- **Pros:** Preserves dataset size (N remains constant).
- **Cons:** Artificially alters the distribution shape.




```
If Value > Upper -> Value = Upper
If Value < Lower -> Value = Lower
```

The Outlier Workflow Summary

Step 1
Check Distribution
Normal or Skewed?

Step 2
Choose Detection
Normal -> Mean +/- 3SD
Skewed -> IQR Method

Step 3
Assess Impact
Weight-Based? (Regression/DL) -> Treat.
Tree-Based? -> Ignore/Verify.

Step 4
Select Treatment
Data Abundance -> Trim.
Data Scarcity -> Cap.

The Data Scientist's Responsibility

Outliers are not inherently "bad". They are signals.

- Signals of Error (e.g., Age 300)
- Signals of Fraud (e.g., Anomaly Detection)
- Signals of Innovation (e.g., The "Einstein" student)

The goal is not just to clean the data, but to understand why the data behaves the way it does.